

Civics Group	Index Number	Name (use BLOCK LETTERS)

H2



**ST. ANDREW'S JUNIOR COLLEGE
2016 JC2 Preliminary Examinations**

H2 BIOLOGY

9648/1

Paper 1: Multiple Choice

Tuesday

20th September 2016

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **40** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

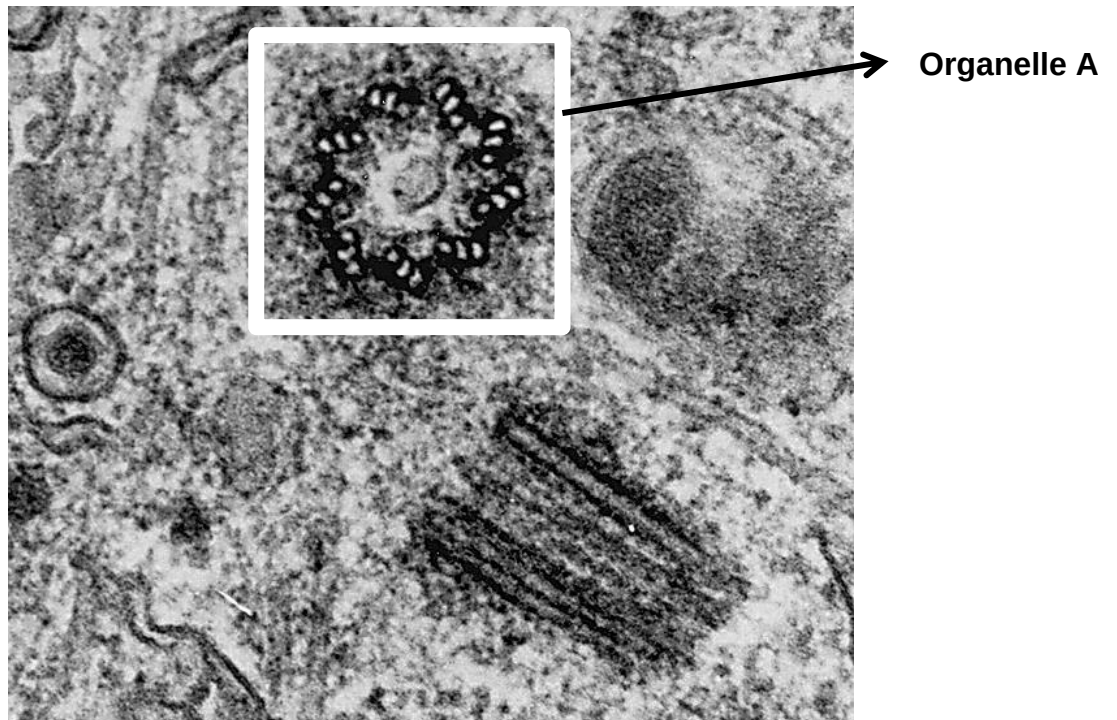
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet only.

This document consists of **27** printed pages.

[Turn over

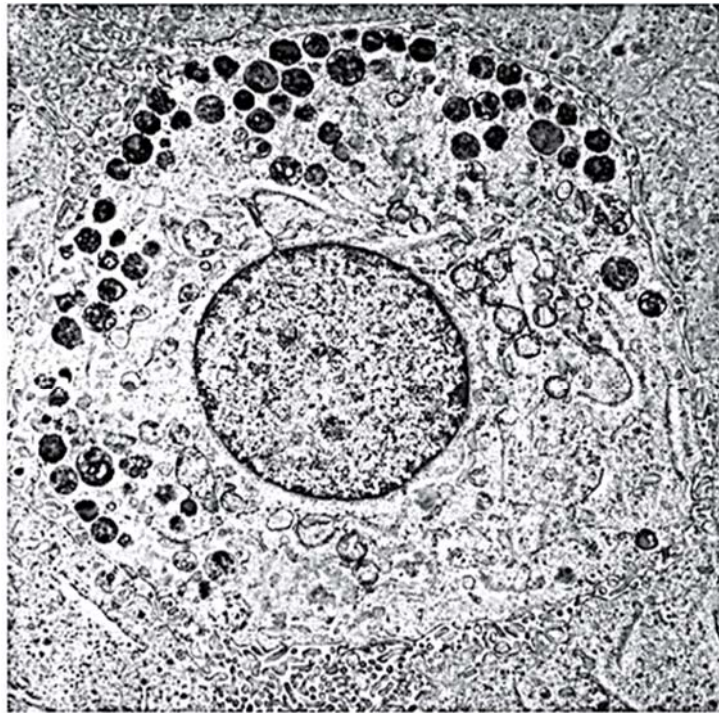
- 1** The figure below shows an electron micrograph of a cross-section of an animal (rat) cell.



Which of the following describes organelle A?

- 1** 9 triplets of microtubules arranged in a ring
 - 2** Inner membrane folded into cristae
 - 3** Synthesises spindle fibres during nuclear division
 - 4** Involved in aerobic respiration
-
- A** 1 only
 - B** 1 and 3 only
 - C** 2 and 3 only
 - D** 2 and 4 only

- 2 The electronmicrograph shows a cell magnified 5700 times.

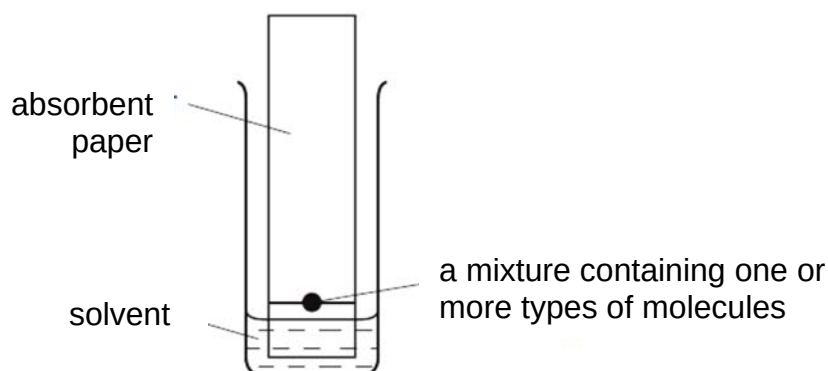


x5700

What is the actual diameter of the nucleus?

- A 0.6 μm
- B 6 μm
- C 35 μm
- D 350 μm

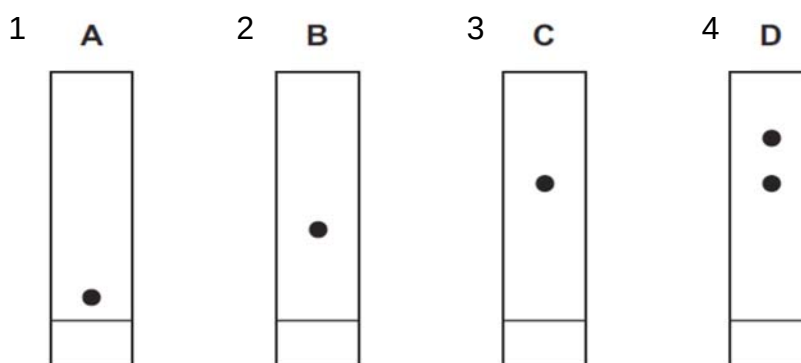
- 3** Chromatography is a technique used to separate molecules by their solubility. The diagram shows an apparatus used for this technique.



As the solvent rises up the paper, the molecules with the greatest solubility in the solvent travel the greatest distances up the paper. When the solvent reaches the top of the paper, the paper is removed, dried and sprayed with a dye. The different molecules appear as coloured spots.

Chromatography was carried out on four different samples – sucrose, cellulose as well as the products of complete hydrolysis of sucrose and cellulose.

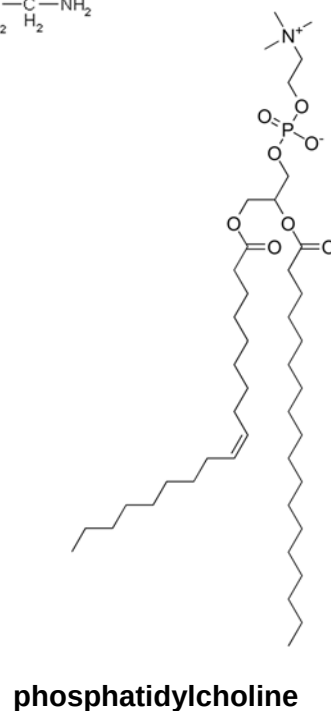
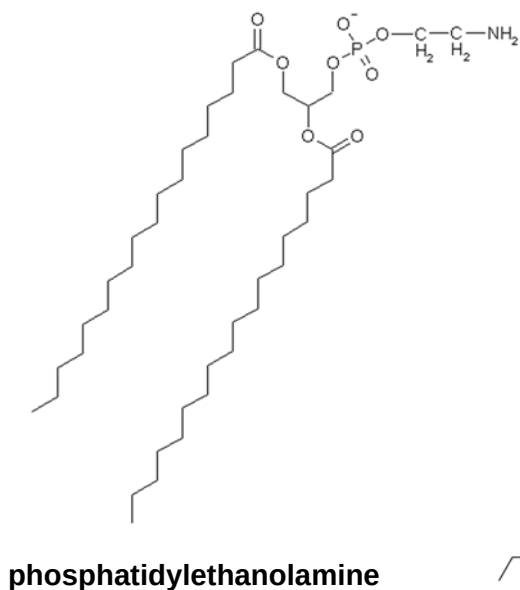
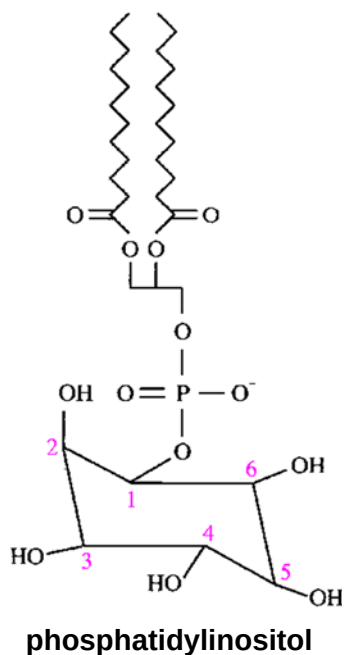
The diagram shows the chromatography results:



Which row shows the correct results?

	sucrose	cellulose	products of complete hydrolysis of sucrose	products of complete hydrolysis of cellulose
A	4	3	1	2
B	3	4	2	1
C	1	2	3	4
D	2	1	4	3

- 4 The diagram below shows three different phospholipids: phosphatidylcholine, phosphatidylethanolamine and phosphatidylinositol.



Which of the following statements is correct?

- A Membranes with a higher phosphatidylcholine content are more fluid.
 B Phosphatidylinositol is unable to form a bilayer in an aqueous environment.
 C Only phosphatidylethanolamine is amphipathic.
 D Only phosphatidylinositol has two fatty acid side chains.
- 5 How many different types of oligopeptides, each composed of 6 amino acids, may be synthesized using the 20 common amino acids?

- A 6^4
 B 20^6
 C 6^{20}
 D 20^4

- 6 During the oxidative phosphorylation process in respiration, energy released from the transport of electrons along the electron transport chain, is used to transport H^+ ions from the mitochondrial matrix to the intermembrane space through membrane proteins.

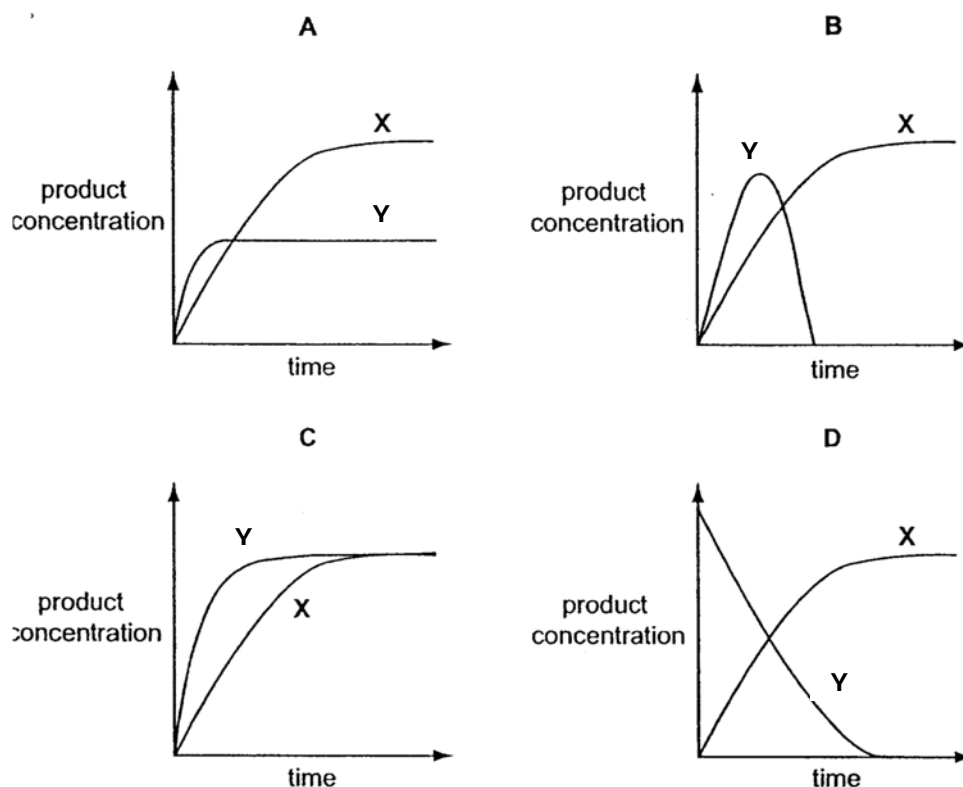
H^+ ions are also transported through ATP synthase from the intermembrane space back into the mitochondrial matrix.

Which of the following shows the correct membrane transport process involved?

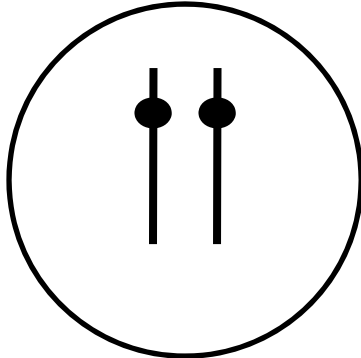
	Transport H^+ ions from the mitochondrial matrix to the intermembrane space.	Transport of H^+ ions from intermembrane space back into the mitochondrial matrix through the ATP synthase
A	Facilitated diffusion	Facilitated diffusion
B	Facilitated diffusion	Active transport
C	Active transport	Facilitated diffusion
D	Active transport	Active transport

- 7 Two enzyme experiments were carried out. The first, experiment X, was carried out at a constant temperature of 37°C . During the second experiment, Y, the temperature was increased from 37°C to 80°C .

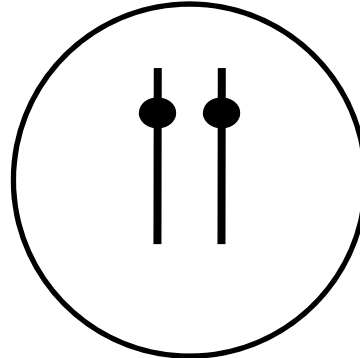
Which graph shows the results?



- 8 S and T are two identical cells of a female. Only the X chromosomes are shown in these cells. When these cells undergo meiosis to form gametes, non-disjunction of the chromosomes occurred. In cell S, it occurred during meiosis I whereas in cell T, it occurred during meiosis II.



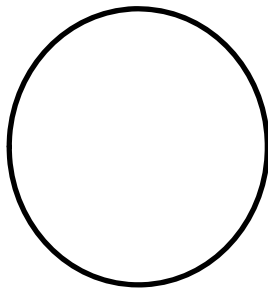
Cell S
(nondisjunction at meiosis I)



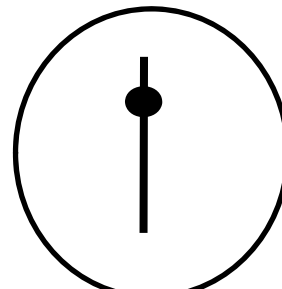
Cell T
(nondisjunction at meiosis II)

The diagrams below show some combination of X chromosome(s) in gametes. Each line represents a non-duplicated chromosome

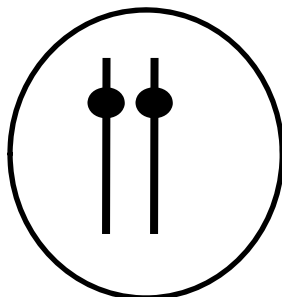
(1)



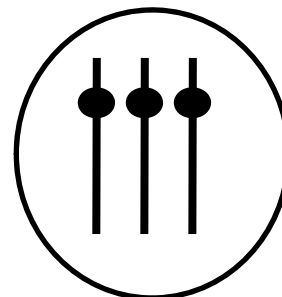
(2)



(3)



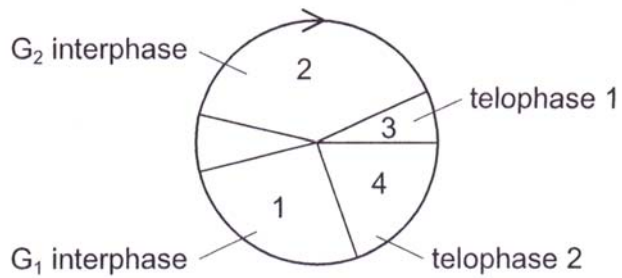
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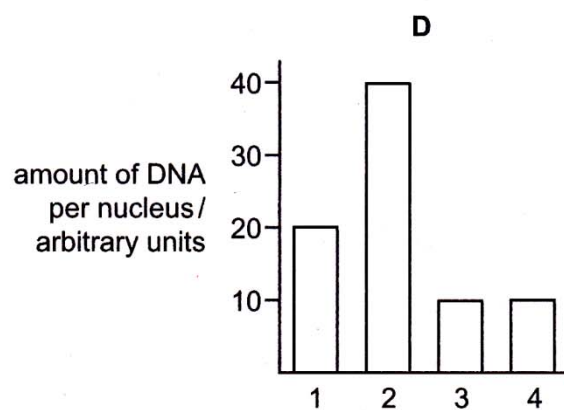
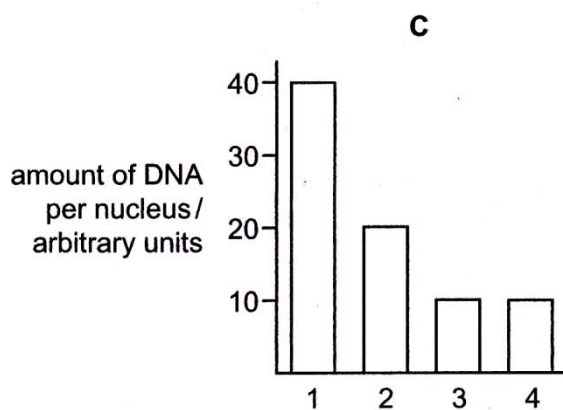
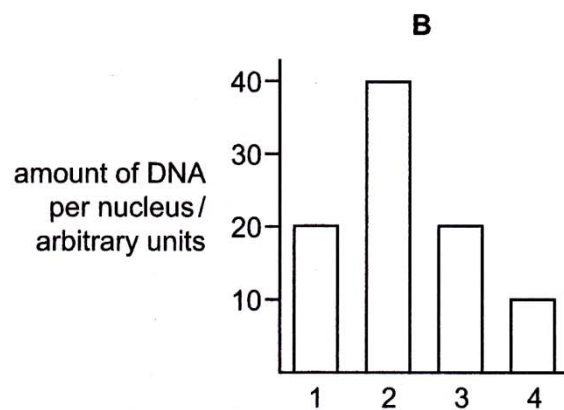
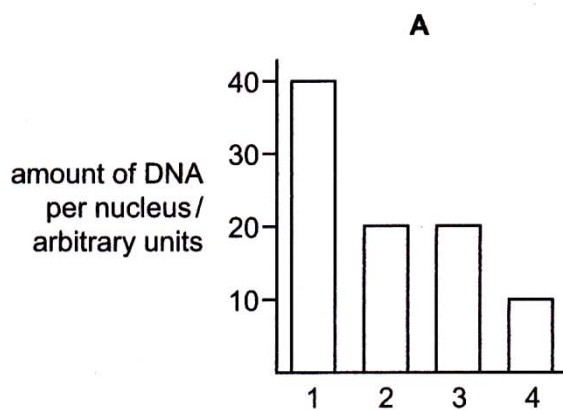
Which of the following options shows correctly the type of gametes that is/are **common** to **both** forms of non-disjunction?

- A 1 only
- B 2 only
- C 1 and 3 only
- D 2 and 4 only

9 The figure below shows the different stages (1 to 4) of a meiotic cell cycle.

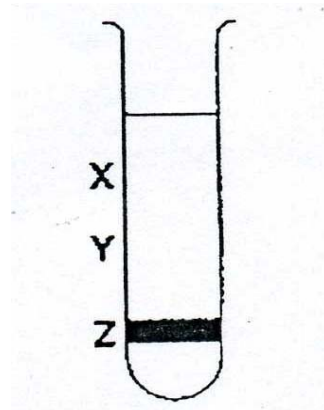


Which bar graph correctly represents the variation in the amount of DNA content per nucleus?



- 10** A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N . The culture was then allowed to reproduce using nucleotides containing normal ^{14}N . The DNA was examined using a centrifuge after one generation and again after two generations.

The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



In which pattern would the DNA be found after the first and after the second cell generations?

	After first generation	After second generation
A	Half at X and half at Y	Quarter at X and at Z and half at Y
B	Half at X and half at Z	Quarter at X and at Z and half at Y
C	All at X	Half at X and half at Y
D	All at Y	Half at X and half at Y

11 The sequence of DNA bases

GGCAATTGGAAACGAATACCCAGT

codes for the following sequence of amino acids.

glycine-asparagine-tryptophan-lysine-arginine-isoleucine-proline-serine

A mutant organism, in which one of the bases was deleted and then inserted at a different point in the DNA molecule, produced a peptide with the following sequence:

glycine-asparagine-tryptophan-asparagine-glutamine-isoleucine-proline-serine

Which base was removed by this mutation?

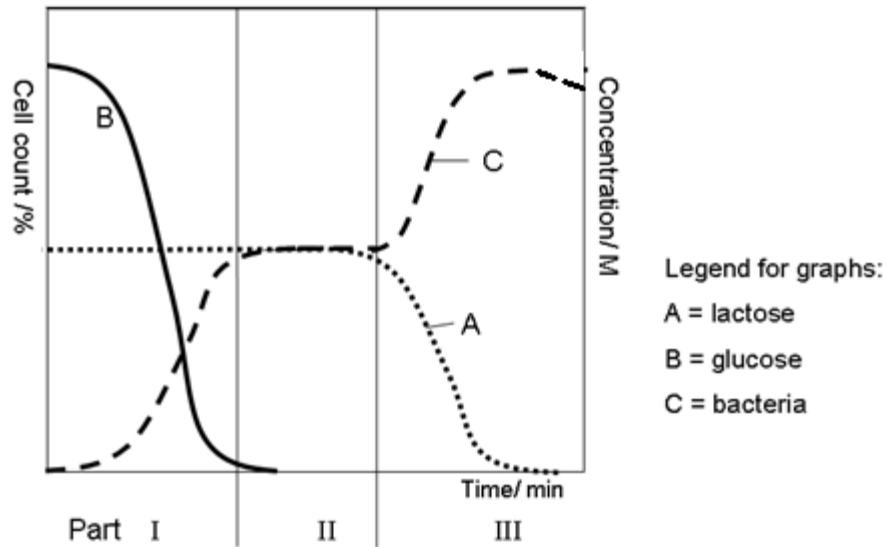
- A** adenine
- B** guanine
- C** cytosine
- D** thymine

12 Which of the following statements about bacterial chromosome structure is/are true?

- 1** Not associated with histone proteins
- 2** Single-stranded chromosome
- 3** Located in the nucleoid region of a nucleus
- 4** Most genes are separated by intergenic DNA sequences

- A** 1 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 2 and 4 only

- 13 The graph below shows the number of bacteria growing in the presence of varying concentrations of glucose and lactose.



Which of the following statements correctly accounts for the bacteria growth at different parts of the graph?

- 1 In Part I, the *lac* repressor is not bound to the operator and the catabolite activator protein (CAP) is not bound to the CAP-binding site, hence expression of structural genes is low.
- 2 In Part II, cAMP levels are high and binding of cAMP to the CAP binding site, hence expression of structural genes is high.
- 3 In Part III, as allolactose levels are depleted, *lac* repressor is active, bacteria growth plateaus.

- A 1 only
 B 1 and 3 only
 C 2 and 3 only
 D 1, 2 and 3

- 14** There are three different types of influenza viruses, types A, B and C. A common vaccine for influenza targets all three of these viral strains, containing inactivated forms of all three viral strains.

However, the vaccine is usually effective only for one year, after which it loses its effectiveness and has to be reformulated. As a result, a new vaccine needs to be administered to an individual every year.

Why do influenza vaccines need to be reformulated every year?

- A** Neuraminidase on the viral envelope may undergo mutation due to lack of proofreading by DNA polymerase.
- B** Antigenic drift may occur where different influenza strains infect the same host cell and exchange genetic material.
- C** Antigenic shift may occur where random mutations result in change in structure of glycoproteins on the viral envelope.
- D** The influenza genome consists of separate RNA segments which allow recombination between different haemagglutinin and neuraminidase genes.

- 15** Influenza virus and Human Immunodeficiency Virus (HIV) differ in their reproductive life cycles.

Which of the following correctly describes the differences in their life cycles?

- 1** Influenza virus enters the host cell via receptor-mediated endocytosis while HIV enters via fusion of HIV envelope with host cell membrane.
- 2** The RNA-dependent RNA polymerase of the influenza virus uses a positive sense RNA strand as the template while the reverse transcriptase of the HIV uses negative sense RNA strand as the template.
- 3** The influenza virus uses the host cell's ribosomes for synthesis of viral proteins while HIV does not.

- A** 1 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 16** The table shows the differences in the average length in kilobases (kb) of a gene and of messenger RNA (mRNA) in a bacterium and a mammal.

organism	average length of gene/kb	average length of mRNA / kb
bacterium	2.0	2.0
mammal	16.6	2.2

Which feature of the mammalian genome does **not** account for these differences?

- A** 94% of mammalian genes are interrupted by introns.
 - B** Mammalian genes have an average of seven introns.
 - C** Initial transcripts of pre-mRNA are shortened by splicing in a mammal.
 - D** The mammalian genome is 1000 times longer than the bacterial genome.
- 17** About 12,000 genes are expressed in both chick liver and oviduct. However an estimated additional 5,000 genes are expressed only in liver, while an additional 3,000 genes are expressed only in oviduct.

Which of the following could explain these observations?

1	The additional genes may have different methylation patterns in different tissues.
2	The concentrations of transcriptional enhancer elements for the additional genes vary in different tissues.
3	The number of genome copies is different in different somatic cells.
4	A common set of genes are expressed for normal functions in liver and oviduct.

- A** 1 only
- B** 2 and 3 only
- C** 1 and 4 only
- D** 2 and 4 only

- 18** A eukaryotic gene of 255 base-pairs codes for 3 different polypeptides of 84 amino acids, 49 amino acids and 35 amino acids.

Which of the following would be possible explanation(s) for the above?

1	Proteolytic cleavage has occurred.
2	Exon skipping has occurred.
3	Polycistronic mRNA contains multiple start and stop codons.
4	Use of alternative polyadenylation signals.

- A** 1 and 2
B 1, 2 and 4
C 1, 3 and 4
D 2, 3 and 4
- 19** Within a cell, the amount of polypeptide made using a given mRNA molecule depends partly on:
- A** the degree of DNA methylation
B the rate at which the mRNA is degraded
C the presence of transcription factors
D the types of ribosomes present in the cytoplasm
- 20** Ras protein is a G-protein involved in a cell-signalling pathway. This protein is coded for by a proto-oncogene. Which of the following would lead to the cell undergoing uncontrolled cell division?
- A** Deletion of the Ras proto-oncogene.
B Duplication of the Ras promoter that controls the proto-oncogene.
C Translocation of the proto-oncogene upstream of a hyperactive promoter.
D Point mutation within the proto-oncogene, forming mutant Ras which is constitutively active.

21 The pedigrees show the inheritance of two genetically transmitted diseases.

Key:



female

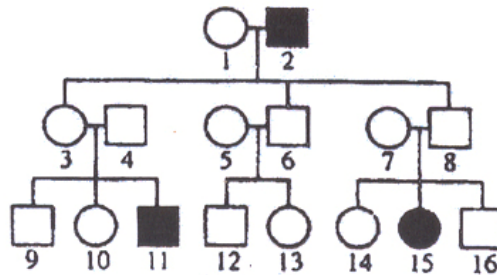


male

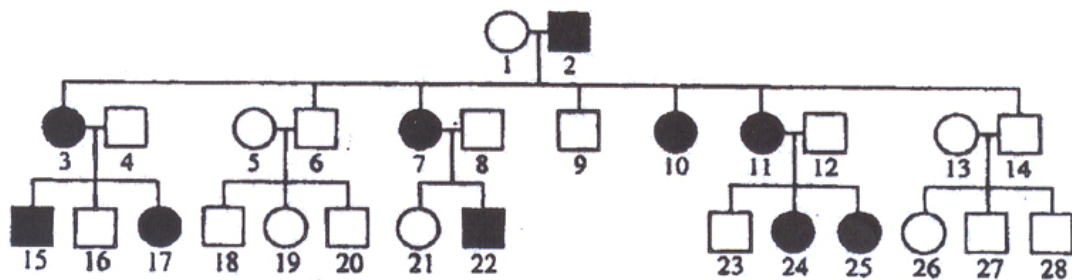


affected individuals

Galactosaemia



Hypophosphataemic rickets



Which row correctly describes the mode of inheritance of galactosaemia and hypophosphataemic rickets?

	Galactosaemia	Hypophosphataemic rickets
A	Autosomal dominant	Sex-linked recessive
B	Autosomal recessive	Sex-linked dominant
C	Sex-linked recessive	Autosomal recessive
D	Sex-linked dominant	Autosomal dominant

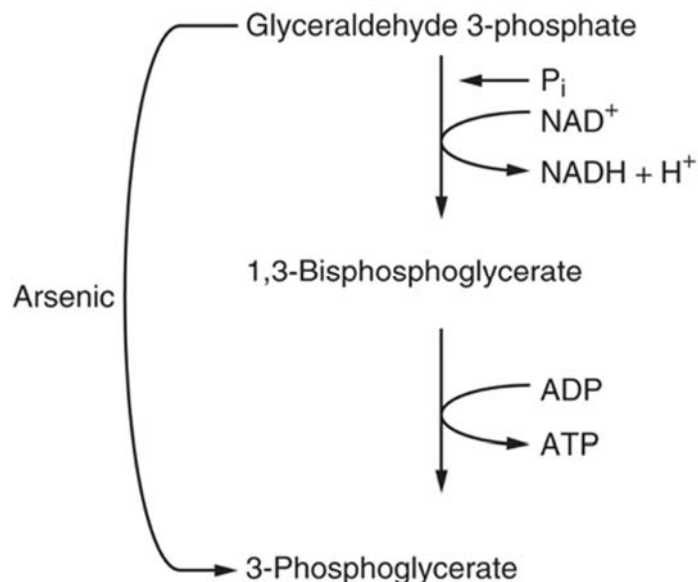
- 22** Some beetles have spines on their legs to help them capture prey. Three alleles determine the presence and type of spines a beetle will produce. The inheritance of a single L^C allele produces curved spines on the legs. The L^S allele is recessive to L^C allele, and produces straight spines. Beetles that are homozygous recessive for the L^n allele produces no spines on its legs.

When a beetle with curved spines was crossed with a beetle with straight spines, they produced the following F_1 generation:

62 curved spines
28 straight spines
30 no spines

If an F_1 offspring with curved spines was picked at random and test crossed with a beetle with no spines, what is the probability of producing beetles with no spines?

- A** $\frac{1}{4}$
B $\frac{1}{2}$
C $\frac{3}{4}$
D 1
- 23** The diagram shows the effect of arsenic on the metabolism of glyceraldehyde-3-phosphate. What is the net yield of ATP molecules from the conversion of one glucose molecule to two pyruvate molecules in the presence of arsenic?



- A** 0
B 1
C 2
D 3

24 Which of the following is true about both cyclic and non-cyclic photophosphorylation?

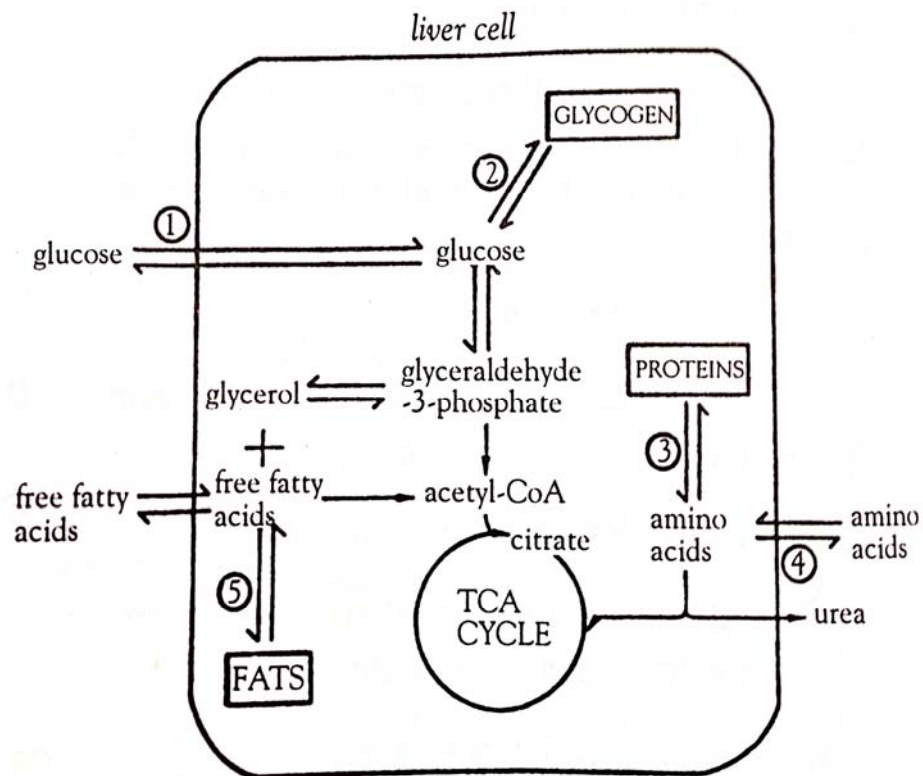
- 1** Establishes an electrochemical gradient across the thylakoid membrane
- 2** Involve photosystem II
- 3** Require oxygen as the final electron acceptor
- 4** Photolysis of water occurs

- A** 1 only
- B** 1 and 2 only
- C** 2 and 4 only
- D** 1, 3 and 4 only

25 Which of the following describes the state of a non-propagating (resting) mammalian axon?

	Concentration in axoplasm		Condition of Na ⁺ /K ⁺ pumps in axon membrane
	Na ⁺	K ⁺	
A	High	Low	Active
B	High	Low	Inactive
C	Low	High	Inactive
D	Low	High	Active

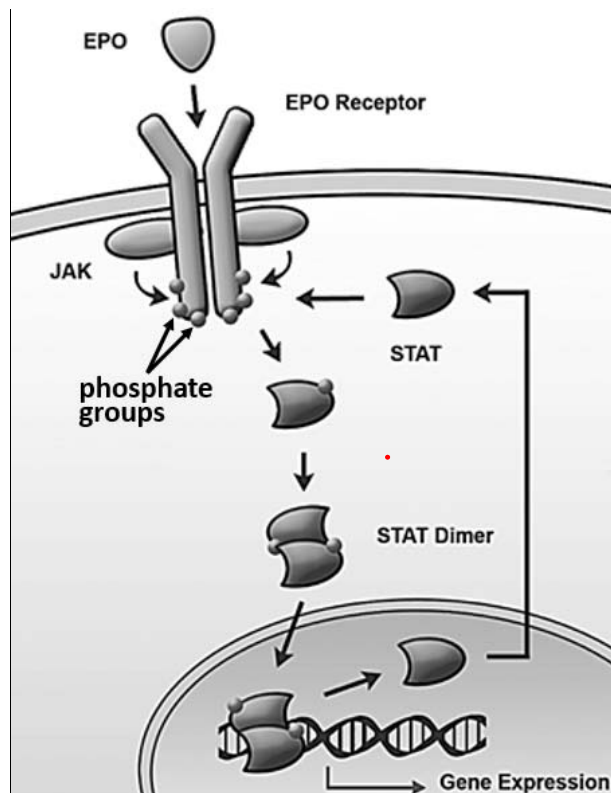
26 The diagram shows some biochemical pathways in a liver cell.



At which numbered points would the hormone insulin accelerate the pathway in the directions indicated?

- A 1, 2 and 3
- B 1, 2 and 5
- C 1, 3 and 4
- D 3, 4 and 5

27 The diagram shows the Jak-STAT cell signalling pathway.

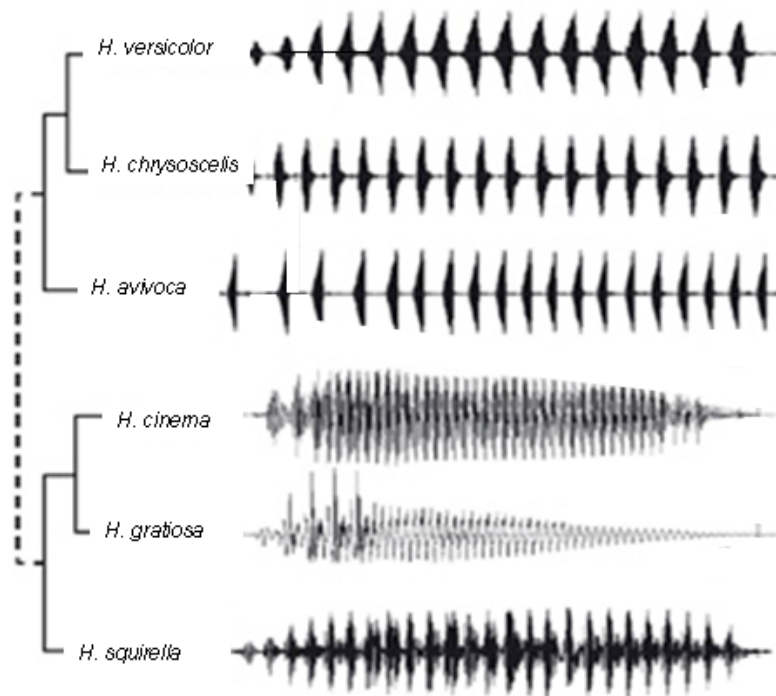


Which of the following statements is/are correct?

- 1 EPO can be a type of steroid hormone.
- 2 Phosphorylation of STAT changes its 3D conformation.
- 3 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.
- 4 Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.

- A 1 and 3 only
- B 2 and 3 only
- C 2 and 4 only
- D 2, 3 and 4 only

- 28 The figure below show the phylogenetic tree of six different species of frogs belonging to the Hyla genus and their mating calls.



Which of the following can be inferred from the chart?

- 1 Frogs with more similar call patterns are more closely related.
 - 2 The frogs are classified based on the duration, intensity, and frequency of their calls.
 - 3 The calls act as a form of isolation mechanism which disrupts gene flow between different species of frogs.
 - 4 Genetic variations exist in the calls among frog populations of each species due to mutations.
- A 1 and 3 only
 B 2 and 3 only
 C 1, 3 and 4 only
 D All of the above

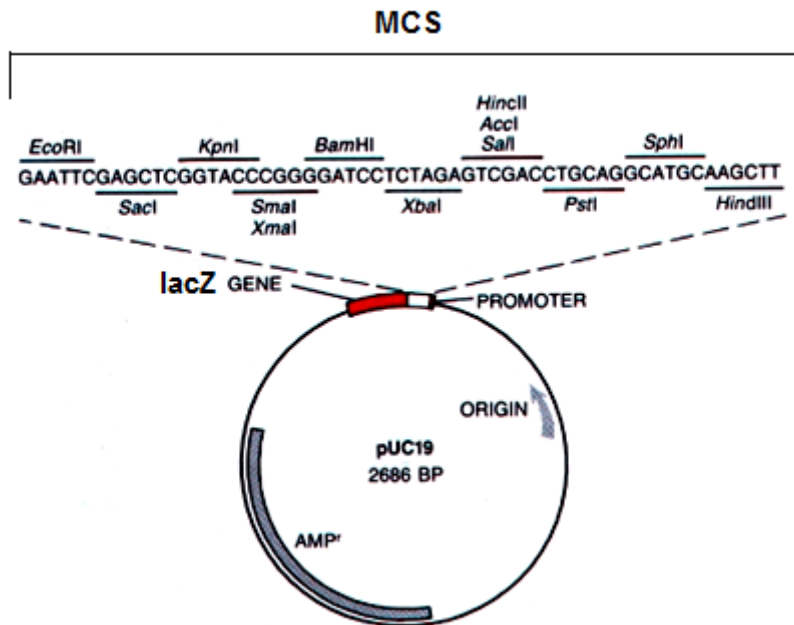
29 Which one of the following statements is related to the principle of adaptive radiation in evolution?

- A** Unrelated organisms living in similar environments may tend to resemble each other.
- B** Species evolve by natural selection, which alters the allele frequencies in interbreeding populations.
- C** Related organisms living in different environments show modifications of an underlying unity of pattern.
- D** Selection in a constant environment tends to maintain the most frequently occurring variety of a species.

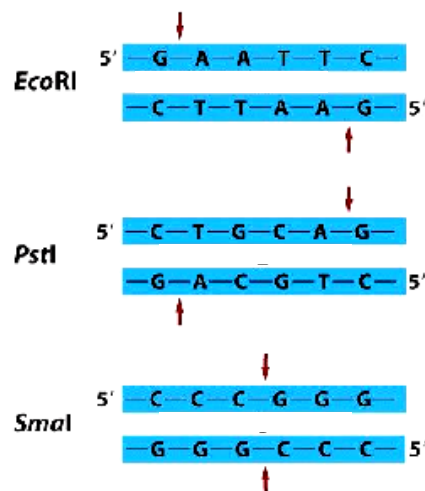
30 Which of the following correctly describes the type of sequences used when investigating closely related and distantly related species?

	Closely related species	Distantly related species
A	DNA sequences which more conserved, such as those found within non-coding sequences	DNA sequences which evolve more quickly, such as those found within coding sequences that codes for proteins that are biologically important for survival.
B	DNA sequences which more conserved, such as those found within coding sequences that codes for proteins that are biologically important for survival	DNA sequences which evolve more quickly, such as those found within non-coding sequences
C	DNA sequences which evolve more quickly, such as those found within coding sequences that codes for proteins that are biologically important for survival.	DNA sequences which are more conserved, such as those found within non-coding sequences
D	DNA sequences which evolve more quickly, such as those found within non-coding sequences	DNA sequences which are more conserved, such as those found within coding sequences that codes for proteins that are biologically important for survival

- 31 The figure shows the plasmid map of pUC19, which includes the positions of the origin of replication (ORI), multiple cloning site (MCS) and two genes – AMP^r and lacZ. pUC19 is used for the production of human anti-thrombin III (hATIII) in the bacterium *E. coli*.



The restriction sites of three particular restriction enzymes, *EcoRI*, *PstI* and *SmaI* are shown below.



Which of the following can be concluded from the figures?

1	It is possible to ensure that the gene is inserted into pUC19 in the correct orientation by using both <i>EcoRI</i> and <i>PstI</i> to cleave pUC19 and the hATIII cDNA.
2	pUC19 cleaved with <i>EcoRI</i> can anneal to hATIII cDNA cleaved with <i>PstI</i> as both restriction enzymes generate single-stranded overhangs of the same length.
3	The use of <i>SmaI</i> will require an additional step of ligating linkers to the ends of pUC19 and hATIII cDNA.
4	Colonies of <i>E. coli</i> transformed with recombinant pUC19 will survive on agar plates containing ampicillin and X-gal, giving a blue appearance.

- A 1 and 3
- B 1 and 4
- C 2 and 3
- D All of the above

32 Human insulin can now be obtained using genetic engineering.

Which of the following are possible explanations to why functional insulin cannot be produced simply by inserting a recombinant plasmid with an insulin cDNA into a bacteria host?

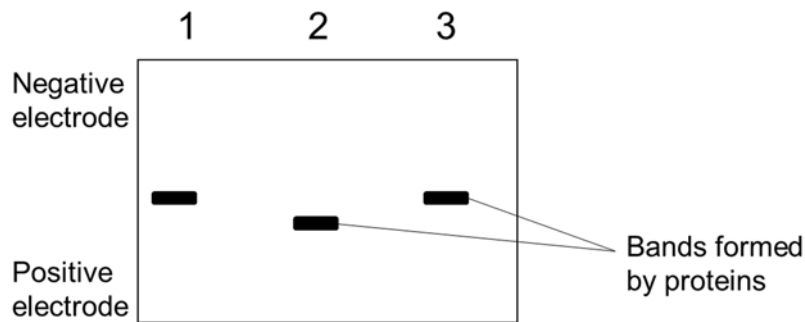
1	The bacteria have no spliceosome.
2	The bacteria will not be able to perform cleavage of polypeptide.
3	The bacteria will not be able to add disulfide bonds to the polypeptide chain.
4	The recombinant plasmid does not contain prokaryotic promoter.

- A 4 only
- B 1 and 2 only
- C 2, 3 and 4 only
- D All of the above

33 Which of the following primers would allow copying of the single stranded DNA sequence 5' ATGGTACCCTAAGTC 3'?

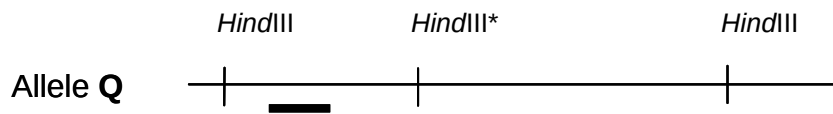
- A 5' ATGGT 3'
- B 5' TACCA 3'
- C 5' CTGAA 3'
- D 5' GACTT 3'

- 34** The diagram shows the results of electrophoresis of the human growth hormone protein extracted from three different individuals.



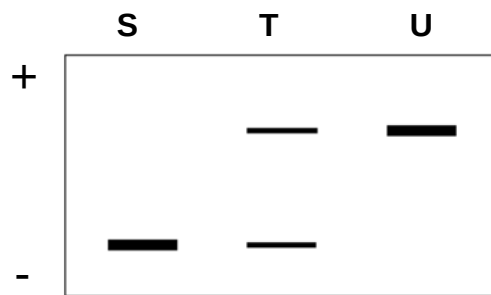
- A mutation occurred in individual 2. Which one of the following can best account for the different banding pattern?
- A** A polar amino acid has been replaced by a non-polar amino acid.
 - B** A polar amino acid has been replaced by a positively-charged amino acid.
 - C** A negatively charged amino acid has been replaced by a non-polar amino acid.
 - D** A non-polar amino acid has been replaced by a negatively-charged amino acid.
- 35** Which of the following poses the greatest limitation of the Human Genome Project in disease analysis?
- A** It is not possible to store the genomic information of all individuals.
 - B** There is too large a genetic difference between individuals to determine the loci of genes of interest.
 - C** Too few model organisms exist for comparative genomic studies to take place.
 - D** Environmental factors and gene interactions in disease are not taken into account.

- 36 The region of the genome containing the RFLP used in this analysis is shown below.



*Hind*III indicates the restriction sites for this enzyme and * indicates the polymorphic site which is missing in the recessive allele **q**. The black bar indicates the position of the probe used to detect the RFLP.

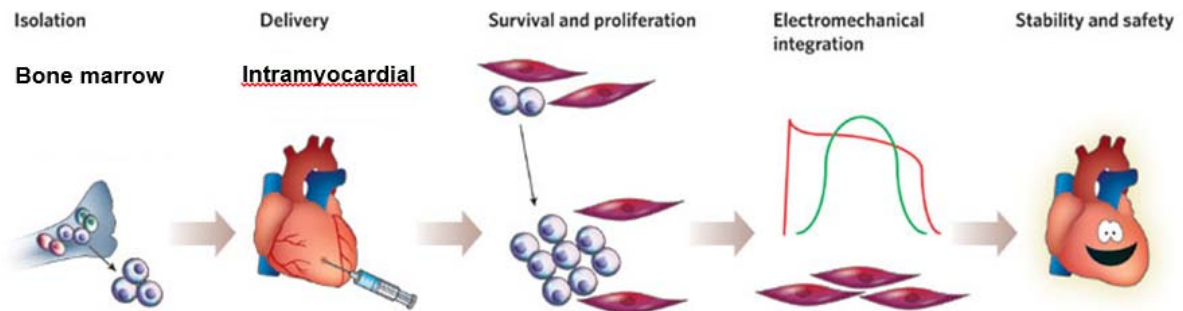
DNA fragments from three different individuals, **S**, **T** and **U**, were subjected to restriction digestion by *Hind*III and separated by gel electrophoresis. The following results were obtained.



Deduce the genotype of individual **S**.

- A QQ
- B X^qY
- C qq
- D X^QY

- 37** Stem cells hold promises for repairing damaged tissues. Recently the use of bone marrow blood stem cells has helped in cardiac repair as shown in the diagram below.



Which of the following conventional notion of stem cells is/are challenged as a result of this experiment?

- 1 The ability of stem cells to self-renew by mitosis.
 - 2 The role of the local tissue environment in stem cell differentiation.
 - 3 The adult stem cells are undifferentiated.
 - 4 The multi-potency of bone marrow blood stem cells.
- A 3 only.
 B 4 only.
 C 3 and 4
 D All of the above
- 38** One therapy to treat β -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β -globin to the patient's cells.

Which of the following would ensure that the normal gene is passed on to the next generation?

- A Using a retrovirus to introduce the mRNA of normal β -globin gene into bone marrow cells.
 B Using a retrovirus to introduce the mRNA of normal β -globin gene into an egg cell.
 C Using an adenoviral vector to introduce the normal β -globin gene into bone marrow cells.
 D Using an adenoviral vector to introduce the normal β -globin gene into an egg cell.

- 39** Which parts of the plant can be used as a choice for an explant in micro-propagation?
- 1** Pollen
 - 2** Root hair
 - 3** Apical bud
 - 4** Flower Petals
- A** 1 only
B 2 and 3 only
C 2, 3 and 4
D All of the above
- 40** In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?
- A** Chloroplasts are surrounded by a double membrane.
B There are no chloroplasts in pollen of most plant species.
C Chloroplasts are smaller than the nucleus.
D There is no DNA in chloroplasts.

END OF PAPER